Project Summary

Project: Customer Support Ticket Intent Classification

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1. Project Overview

This project involved manually annotating a dataset of customer support messages by assigning each message to its primary intent category. The project was designed to simulate a real-world AI data annotation workflow commonly used in Natural Language Processing (NLP) applications, including customer support automation, ticket routing, and chatbot training.

The complete workflow included dataset annotation, quality assurance review, documentation, and project reporting.

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2. Project Objectives

The objectives of this project were to:

- Apply consistent intent classification across a customer support dataset.
- Follow predefined annotation guidelines.
- Improve annotation quality through manual QA review.
- Document the annotation workflow and decision-making process.
- Build practical experience in AI data annotation.

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3. Dataset Information

Item	**Description**
Dataset Size	100 customer support messages
Language	English
Annotation Type	Single-label text classification
Number of Labels	5
Annotation Method	Manual

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4. Intent Categories

The dataset was annotated using the following intent labels:

- Billing
- Account
- Shipping
- Technical Issue
- General Inquiry

Each message received one label representing the customer's primary intent.

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5. Annotation Workflow

The project followed a structured annotation process:

1. Review annotation guidelines.
2. Read each customer message.
3. Determine the customer's primary intent.
4. Assign the appropriate label.
5. Record notes for ambiguous cases when necessary.
6. Conduct a manual quality assurance review.
7. Resolve inconsistencies.
8. Finalize the annotated dataset.

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6. Skills Demonstrated

This project demonstrates practical experience in:

- AI Data Annotation
- Text Classification
- Intent Classification
- Data Labeling
- Annotation Quality Assurance (QA)
- Annotation Documentation

- Critical Thinking
- Analytical Reasoning
- Natural Language Processing (NLP)

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7. Tools Used

- Microsoft Excel
- Microsoft Word
- GitHub
- ChatGPT (used to simulate annotation tasks, generate representative datasets, and support QA discussions)

Transparency Note: This project is part of a self-directed portfolio created to demonstrate annotation workflows. ChatGPT was used as a learning and collaboration tool during project development, while annotation decisions, reviews, and documentation were completed manually.

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8. Key Outcomes

Upon completion of this project, I was able to:

- Build and annotate a structured intent classification dataset.
- Apply annotation guidelines consistently across multiple samples.
- Review and improve annotation quality through QA.
- Document annotation decisions and project methodology.

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9. Future Improvements

Potential enhancements for this project include:

- Expanding the dataset beyond 100 samples.
- Introducing additional intent categories.
- Measuring inter-annotator agreement using multiple reviewers.
- Exploring semi-automated annotation workflows.